Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/13/2005
PI Name:	Thomas, John Ph.D.		
Project Title:	Quantum Coherence in Ultracold Fermionic Vapors		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27708	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	10/01/2001	**End Date:**	04/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNCO4GA14G		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	The purpose of this program is a fundamental study of coherence between ground state hyperfine levels in an ultracold vapor of lithium-6 fermions contained in an ultrastable CO2 laser trap. This system permits important proof-of-principle experiments which exploit the nointeracting properties of a single component Fermi gas. The objective of this program is to broadly investigate coherence between ground hyperfine states in an ultracold fermionic gas of lithium-6 contained in an ultrastable all-optical trap. For fermions, the Pauli exclusion principle provides unique opportunities for increasing the spectroscopic resolultion. Proposed experiments include measurement of collisional shifts to determine the magnitude and sign of elastic scattering amplitudes in two-component mixtures, demonstration of double suppression of shifts and inelastic rates for coherent superposition states, demonstration of super-long coherence lifetimes for superposition states fo fermions in an ultrastable optical trap, and potentially measurement of frequency shifts arising from Cooper pairing. The ultimate coherence lifetime is expected to be limited by tthe gravitational potential and the		

	Pauli exclusion principle. The significance of the proposed program is two-fold. First, the program explores new methods for suppressing interactions by using fermionic atoms. This may impact the development of sensitive high resolution spectroscopic methods for precision measurement and atomic clocks. Further, the program utilizes and ultrastable all-optical trap with an exceptionally long lifetime and extremely low residual heating rates. This system will enable storage and study of diamagnetic atoms and molecules which could not otherwise be trapped. Second, the lithium-6 system exhibits magnetically tunable pairing interactions and is therefore an ultracold gas analog of a superconductor. Hence, it will enable precision tests of superconductivity theories in systems with widely tunable interactions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research on strongly-interacting Fermi gases impacts the theory of super-high temperature superconductors, which one day may enable lossless communication, power distribution, and magnetic levitators for transportation.
Task Progress:	Optically- trapped, strongly- interacting Fermi gases provide a paradigm for strong interactions in nature, and are therefore of interest in fields well outside of atomic physics. These include studies of neutron stars and nuclear matter, quark-gluon plasmas, and high temperature superconductors. Strong interactions are produced in atomic gases by using a magnetic field to tune near a Feshbach resonance, which enhances the two-body scattering amplitude. In December of 2004, we achieved a major breakthrough in our observations of a strongly-interacting degenerate Fermi gas, which we first produced in 2002. We measured for the first time the heat capacity of the gas, and observed a phase transition, which was interpreted as the onset of high-temperature superfluidity by our theoretical colleagues at the University of Chicago. A paper reporting this result was published on January 27 in Science Express Online and will appear in Science Magazine. Heat capacity measurement is simple in principle: Energy is added to the system and the temperature is measured. However, to accomplish these measurements in a strongly-interacting Fermi gas, our group had to develop two new experimental techniques, a method for adding a known energy and a method for direct thermometry. Based on the universal properties of the gas, we were able to devise a simple method for precise energy input. We based our direct thermometry on the shape of the cloud. Using these methods, we were able to measure an empirical temperature as a function of the total energy of the gas. Our theoretical colleagues provided the guidance needed to calibrate the empirical temperature by predicting, for the first time in a single theory, both the energy of the gas and the shape of the cloud as a function of temperature. By applying our direct thermometry method to the theoretical cloud profiles, we were able to calibrate the empirical temperature to the theoretical temperature. The theory of the energy is in excellent agreement with the data, and predicts the onset of high-temperature superfluidity at the observed transition temperature. Remarkably, the transition occurs at 30% of the Fermi temperature of the gas. Although the Fermi temperature of the gas is a mere 2.5 microkelvin, in a condensed matter system, the corresponding Fermi temperature would be about ten thousand degrees. Hence, if a condensed matter system can be made to mimic such a strongly-interacting Fermi gas, the corresponding superfluid transition would occur at thousands of degrees, corresponding to a super-high temperature superconductor. In addition to this result, we also published a PRA Rapid Communication on the precision measurement of the breathing mode frequency as a function of magnetic field. The results test the best current theories of many-body dynamics in strongly-interacting matter by exploring the predictions for the equation of state. The data is in qualitative agreement with the theory and in quantitative agreement near the Feshbach resonance. However, the theory breaks down at magnetic fields well above the resonance, for reasons which are not yet understood. Finally, we have measured the temperature dependence of the breathing mode frequency and damping rate. The results reveal a transition in behavior at a temperature consistent with our heat capacity measurements. Together, the heat capacity and damping measurements provide the first determination of the superfluid transition temperature in a strongly-interacting Fermi gas.
Bibliography Type:	Description: (Last Updated:)
Articles in Peer-reviewed Journals	Kinast J, Turlapov A, Thomas JE, Chen Q, Stajic J, Levin K. "Heat capacity of a strongly-interacting Fermi gas." Science. 2005 Feb 25;307(5713):1296-9. PMID: 15681340 , Feb-2005
Articles in Peer-reviewed Journals	Kinast J, Turlapov A, Thomas JE. "Damping of a unitary Fermi gas." Phys Rev Lett. 2005 May 6;94(17):170404. PMID: 15904273 , May-2005
Articles in Peer-reviewed Journals	Kinast J, Turlapov A, Thomas JE. "Breakdown of hydrodynamics in the radial breathing mode of a strongly-interacting Fermi gas." Phys Rev A. 2004 Nov;70: 051401. , Nov-2004